

# Year 3 Maths — Sample Course Content

Sample material for Puzzle Kingdom development use only. Content follows the DfE National Curriculum for Year 3 (England, including London — the curriculum does not vary by city). Not an official publication.

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## 1. The 2, 5 and 10 Times Tables

Multiplying a number means adding equal groups of it together.  $3 \times 5$  means “3 groups of 5”, which is the same as  $5 + 5 + 5 = 15$ .

### The 2 times table

2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24 — every answer is an even number.

### The 5 times table

5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60 — every answer ends in 0 or 5.

### The 10 times table

10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120 — every answer ends in 0.

**Worked example:** A tricycle has 3 wheels. How many wheels do 5 tricycles have?  
 $5 \text{ tricycles} \times 3 \text{ wheels} = 15 \text{ wheels}$ . We can also think of it as the 3 times table: 3, 6, 9, 12, **15**.

## Practice questions

- 1  $7 \times 2 = ?$
- 2  $9 \times 5 = ?$
- 3  $6 \times 10 = ?$
- 4 A spider has 8 legs. How many legs do 2 spiders have?
- 5 Sam has 4 bags with 5 marbles in each bag. How many marbles does Sam have in total?

## 2. Place Value — Hundreds, Tens and Ones

In a 3-digit number, each digit has a value depending on its position: hundreds (H), tens (T), and ones (O). 10 tens are equivalent to 1 hundred — 100 is ten times the size of 10.

| Hundreds | Tens | Ones | Number |
|----------|------|------|--------|
| 3        | 4    | 7    | 347    |
| 2        | 0    | 5    | 205    |
| 6        | 1    | 0    | 610    |

**Worked example:** Partition 583 into hundreds, tens and ones.

$583 = 500 + 80 + 3 = 5 \text{ hundreds} + 8 \text{ tens} + 3 \text{ ones}$ .

We can also partition it in other ways:  $583 = 400 + 180 + 3$ , or  $583 = 500 + 70 + 13$ .

### Practice questions

- 1 What is the value of the digit 6 in the number 462?
- 2 Write 709 in expanded form (hundreds + tens + ones).
- 3 Which is greater: 358 or 385?
- 4 What number is 10 more than 274?
- 5 Order these numbers from smallest to largest: 342, 234, 423, 243.

### 3. Unit Fractions

A unit fraction is one equal part of a whole, such as one half ( $\frac{1}{2}$ ), one quarter ( $\frac{1}{4}$ ), or one fifth ( $\frac{1}{5}$ ). The bottom number (denominator) tells us how many equal parts the whole has been split into.

**Worked example:** Find  $\frac{1}{4}$  of 12 sweets.

Split 12 sweets into 4 equal groups:  $12 \div 4 = 3$ . So  $\frac{1}{4}$  of 12 is 3.

**Worked example:** Find  $\frac{1}{5}$  of 20 stickers.

Split 20 stickers into 5 equal groups:  $20 \div 5 = 4$ . So  $\frac{1}{5}$  of 20 is 4.

#### Practice questions

- 1 Find  $\frac{1}{2}$  of 16.
- 2 Find  $\frac{1}{4}$  of 24.
- 3 Find  $\frac{1}{10}$  of 50.
- 4 A pizza is cut into 8 equal slices. What fraction is 1 slice?
- 5 Priya has 15 stickers and gives  $\frac{1}{3}$  of them to her friend. How many stickers does she give away?